

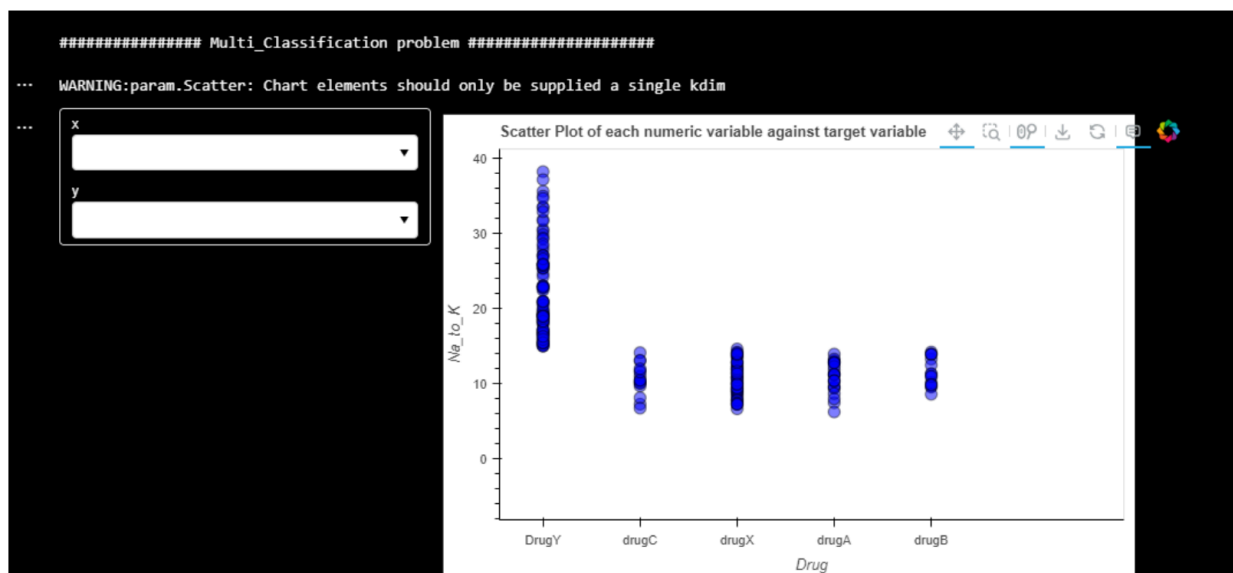
Let's first talk about **classification** using **AutoViz**, **Dataprep**, **Phik Matrix**, **Sweetviz**, and **YData Profiling**. I will briefly explain the general purpose of each library, then show where I used it, along with images and descriptions. I will also highlight the main features of each library.

Here I am getting an example Classification with each library after explanation of Classification then I will take Regression as well.

There are image folders, each image folder is named with the drug, its mean its related to classification. If it has the name of insurance, then its mean it is related to regression.

1-AutoViz is used to automatically generate visualizations and basic data analysis from a dataset with minimal code.

Its main purpose is to quickly explore, understand patterns, relationships, and distributions in data.

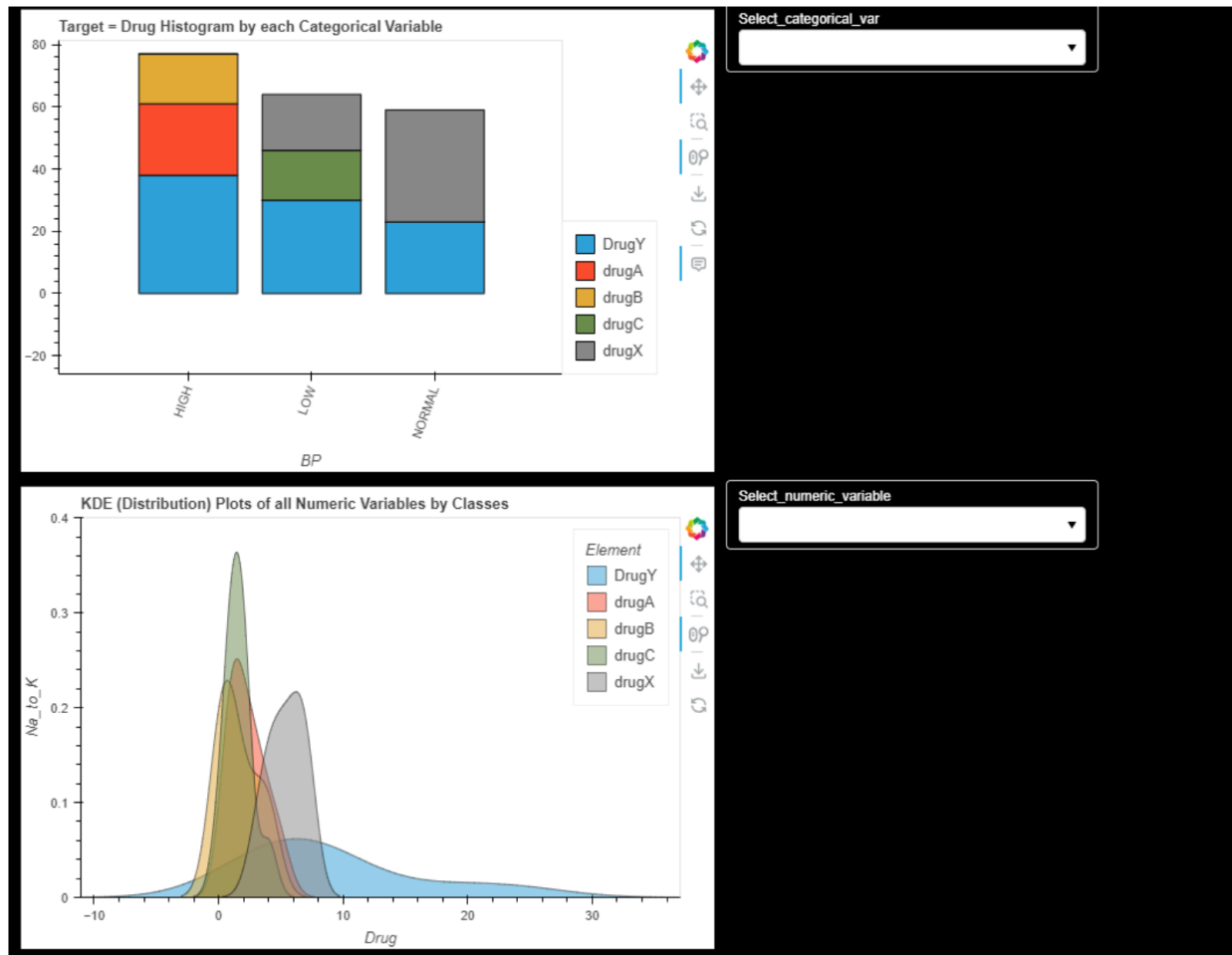


Scatter Plot (Numeric vs Target):

Displays the numeric variable Na_to_K plotted against the target variable Drug (DrugY, DrugX, DrugA, DrugB, DrugC).

The scatter shows distinct clusters for each drug type — DrugY and DrugX tend to have higher Na_to_K ratios, while DrugA, B, and C remain lower.

Na_to_K appears to be a strong differentiating feature useful for predicting drug type in a multi-class classification problem.



Target Drug Histogram by Categorical Variable:

Represents the distribution of Drug classes across the categorical variable BP (Blood Pressure: HIGH, LOW, NORMAL) using stacked bars.

Each color corresponds to a specific drug type, revealing how drugs vary across BP categories.

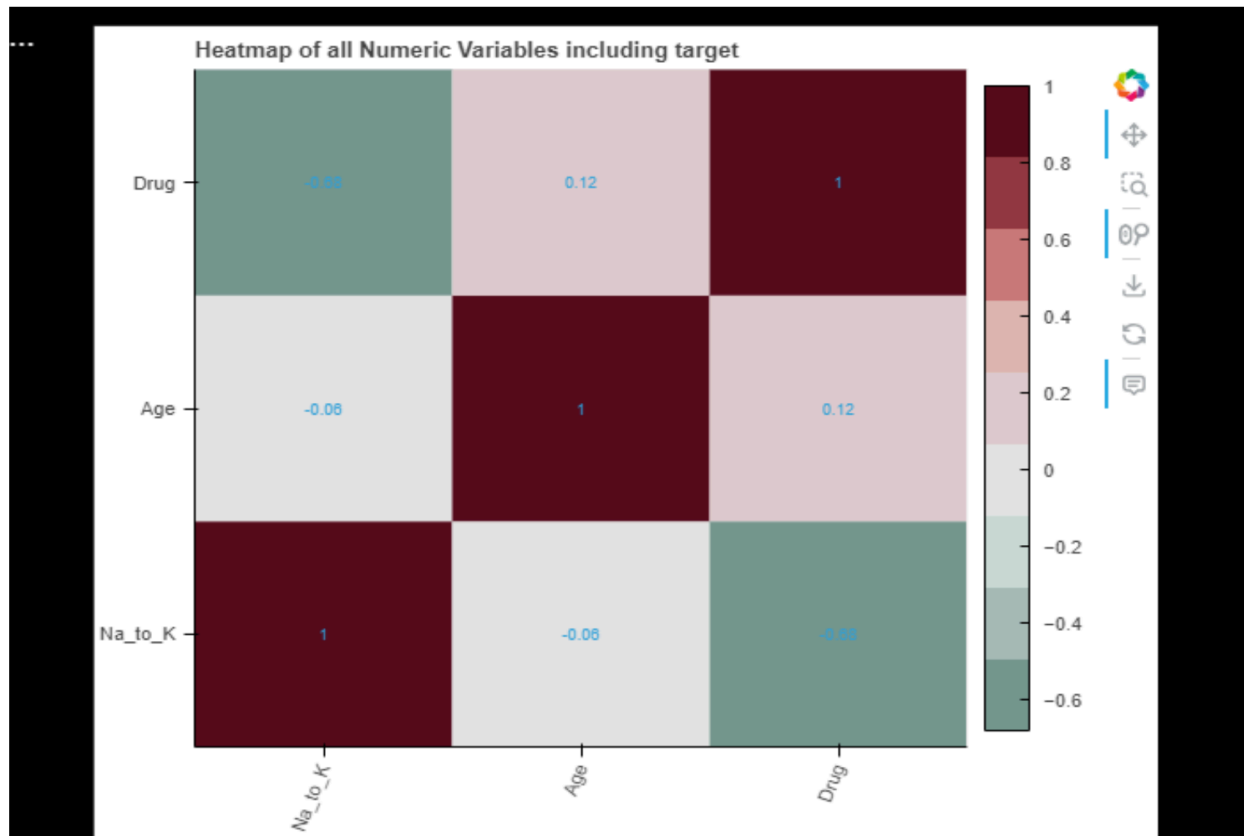
Shows that categorical variables like BP have a significant impact on drug type distribution, highlighting relationships between categorical features and target class.

KDE (Kernel Density Estimation) Plot:

Displays the density distribution of the numeric variable Na_to_K across all drug classes.

Overlapping curves reveal how each drug type differs in numeric distribution—DrugY has a wider spread, while others cluster in narrow ranges.

Demonstrates that Na_to_K provides clear separability among drug classes, supporting its use for multi-class prediction modeling.



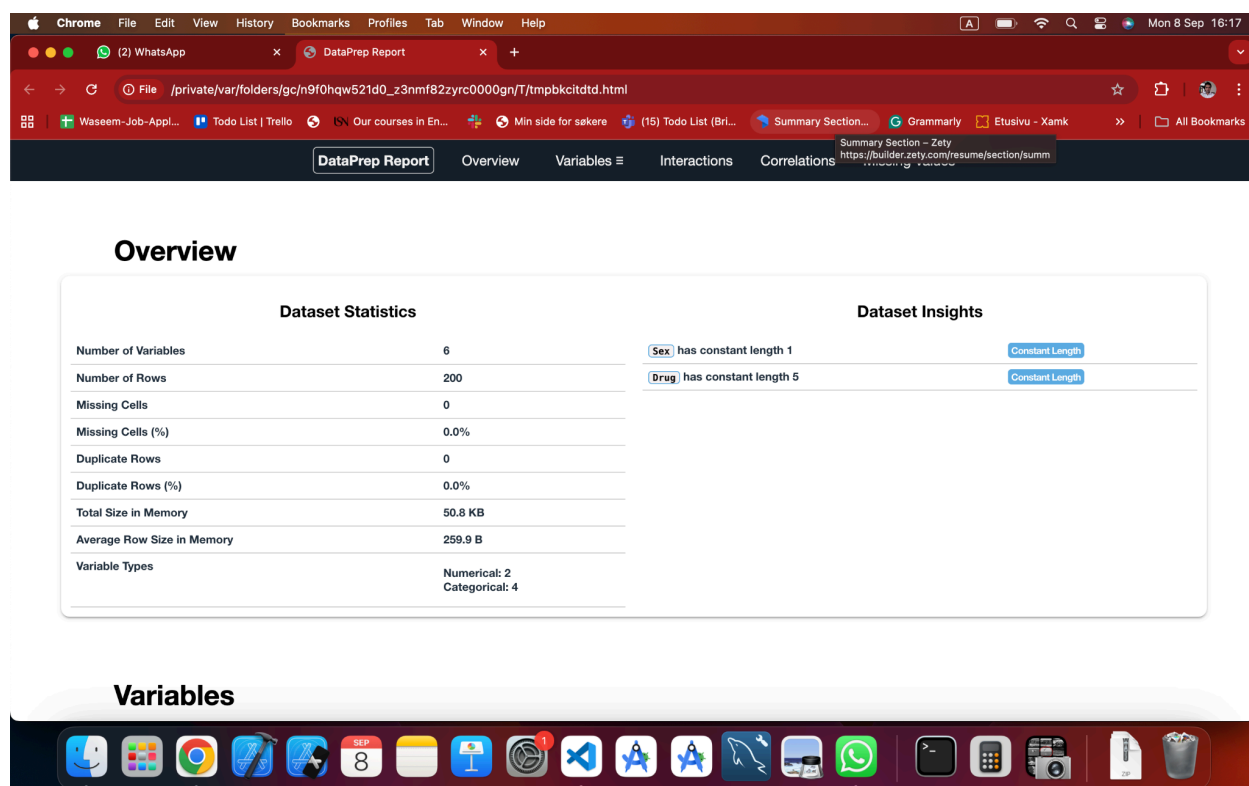
Heatmap of all Numeric Variables, including target, shows the correlation between the numeric variables Drug, Age, and Na_to_K. The correlation between Na_to_K and Drug is -0.06, between Age and Drug is 0.12, and between Age and Na_to_K is also -0.06. These low correlation values indicate that none of the numeric variables have a strong linear relationship with each other. The main point is that each variable provides unique and independent information useful for multi-class classification.

Dataprep

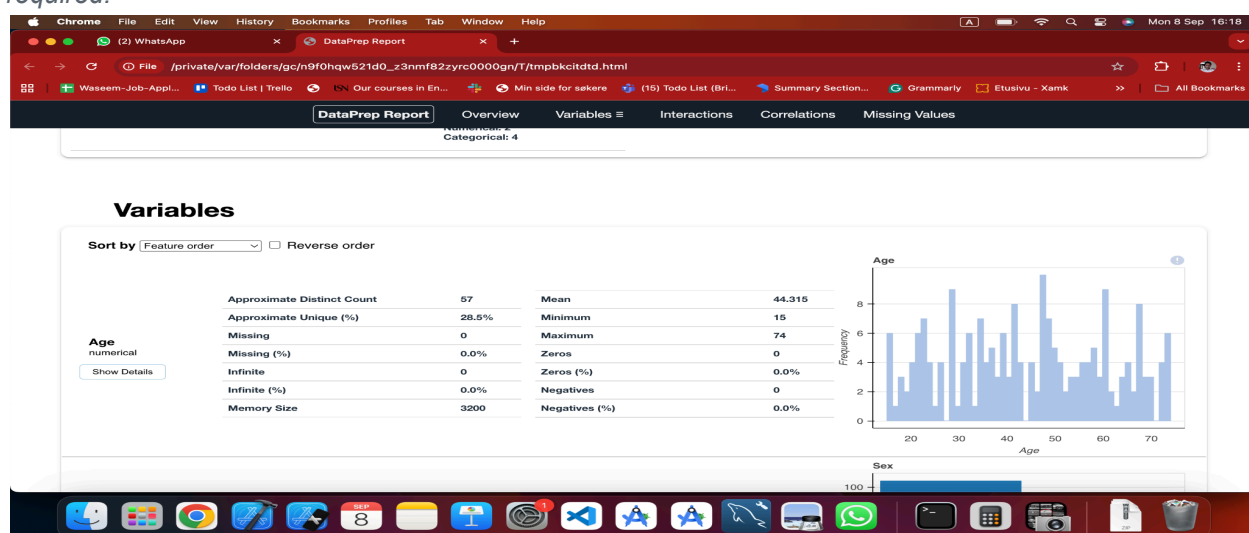
The **DataPrep EDA** library is used to **automatically generate an exploratory data analysis (EDA) report**. Its main purpose is to **quickly summarize, visualize, and detect patterns or issues in a dataset**.

Its another library purpose would be the same, but AutoViz shows charts to understand data, while DataPrep EDA gives a full, simple report explaining the data.

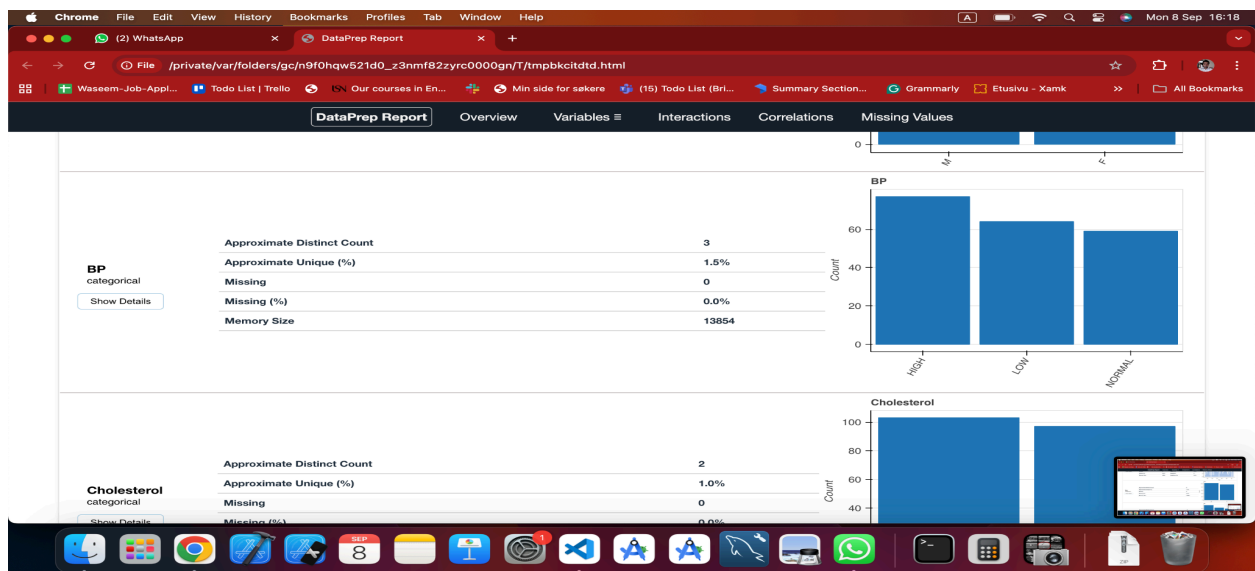
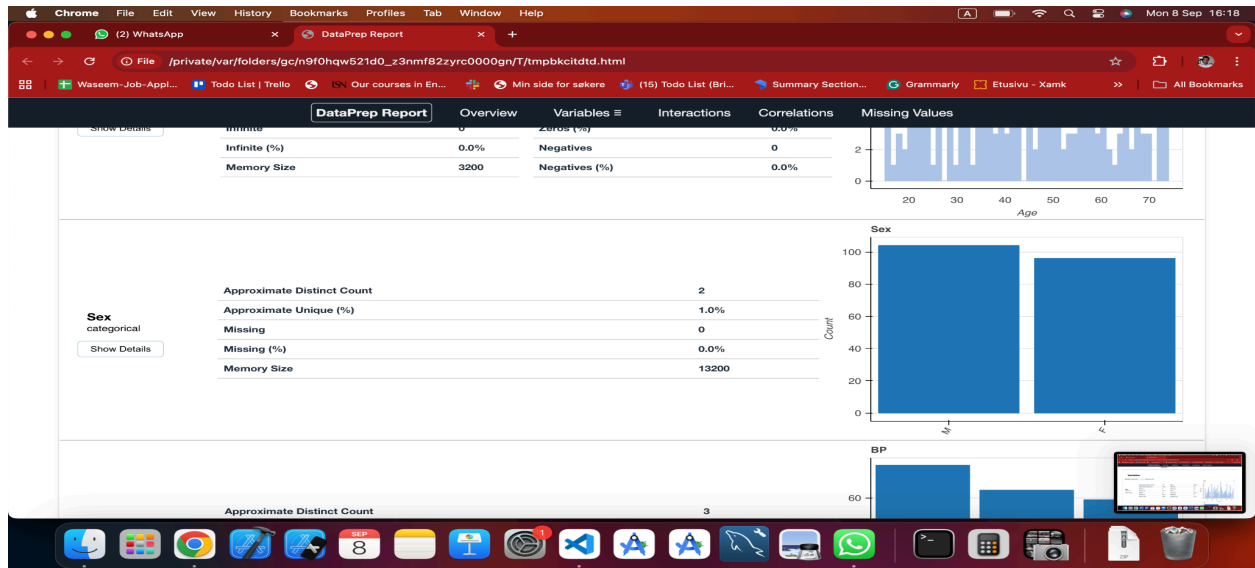
Let's see the report and images with descriptions as well.



The dataset contains 200 rows and 6 variables. All data is complete, with no missing or duplicate entries. Memory usage is low at just 50.8 KB, indicating it's lightweight and quick to load. Of the variables, two are numerical, and four are categorical, making it a mixed-type dataset suitable for both statistical and machine learning tasks. The variables "Sex" and "Drug" have consistent string lengths, which suggests standardized encoding or labeling. Overall, the data quality appears high, with minimal preprocessing required.



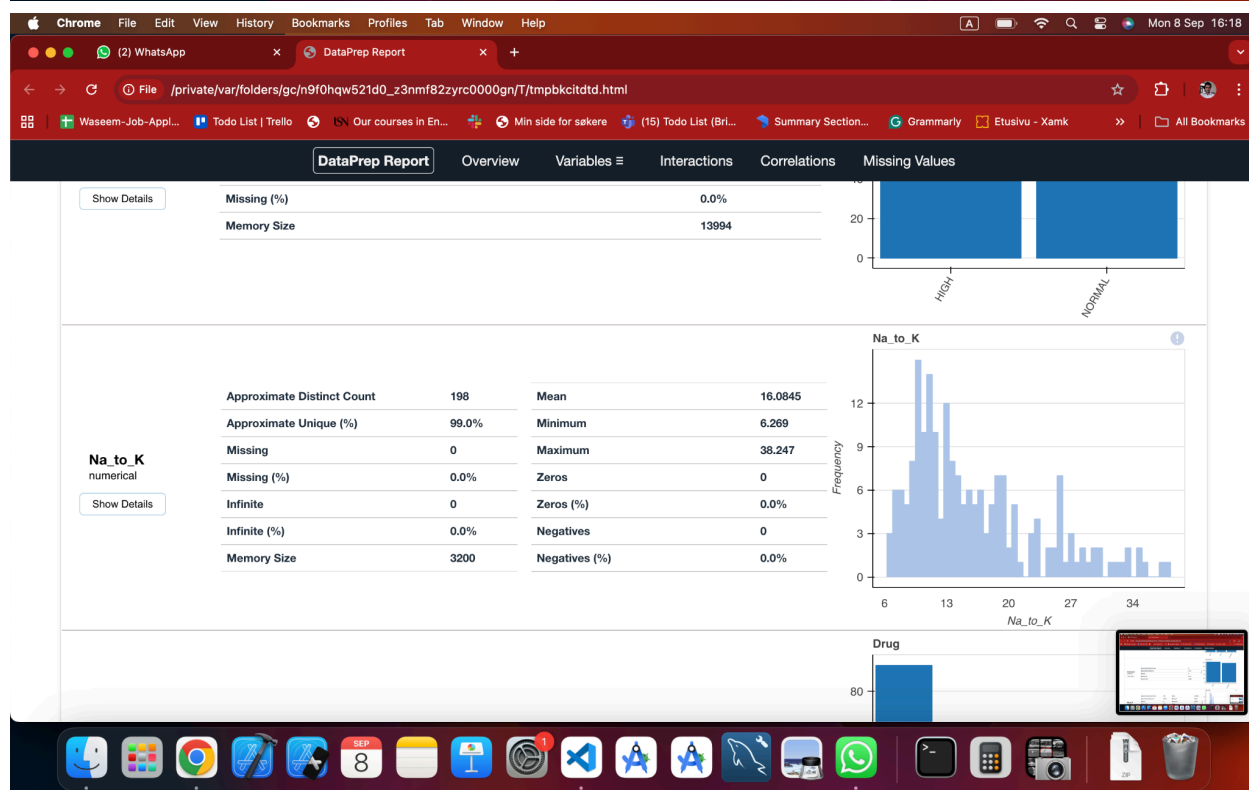
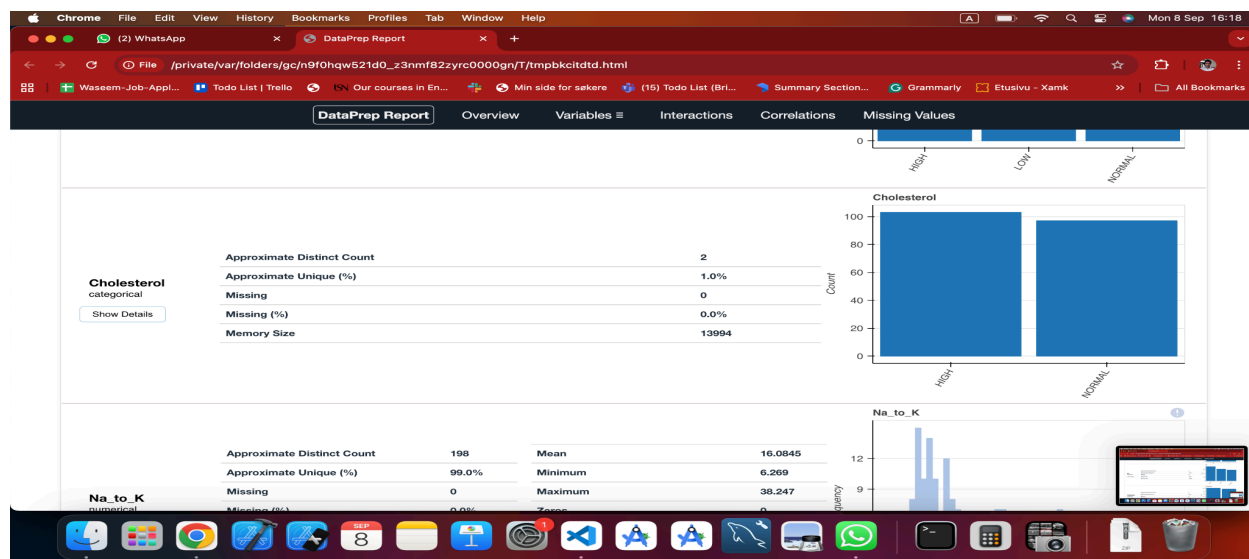
The Age variable ranges from 15 to 74, with a mean of 44.315. It has 57 distinct values (28.5% unique) and no missing, zero, or negative values. The data is clean and evenly distributed across age groups.



SEX and BP

Sex has 2 distinct categories with 1.0% approximate uniqueness and 0.0% missing data, using 13,200 bytes of memory.

BP has 3 distinct categories with 1.5% approximate uniqueness, no missing data, and uses 13,854 bytes. Both variables are clean and well-distributed across their categories.

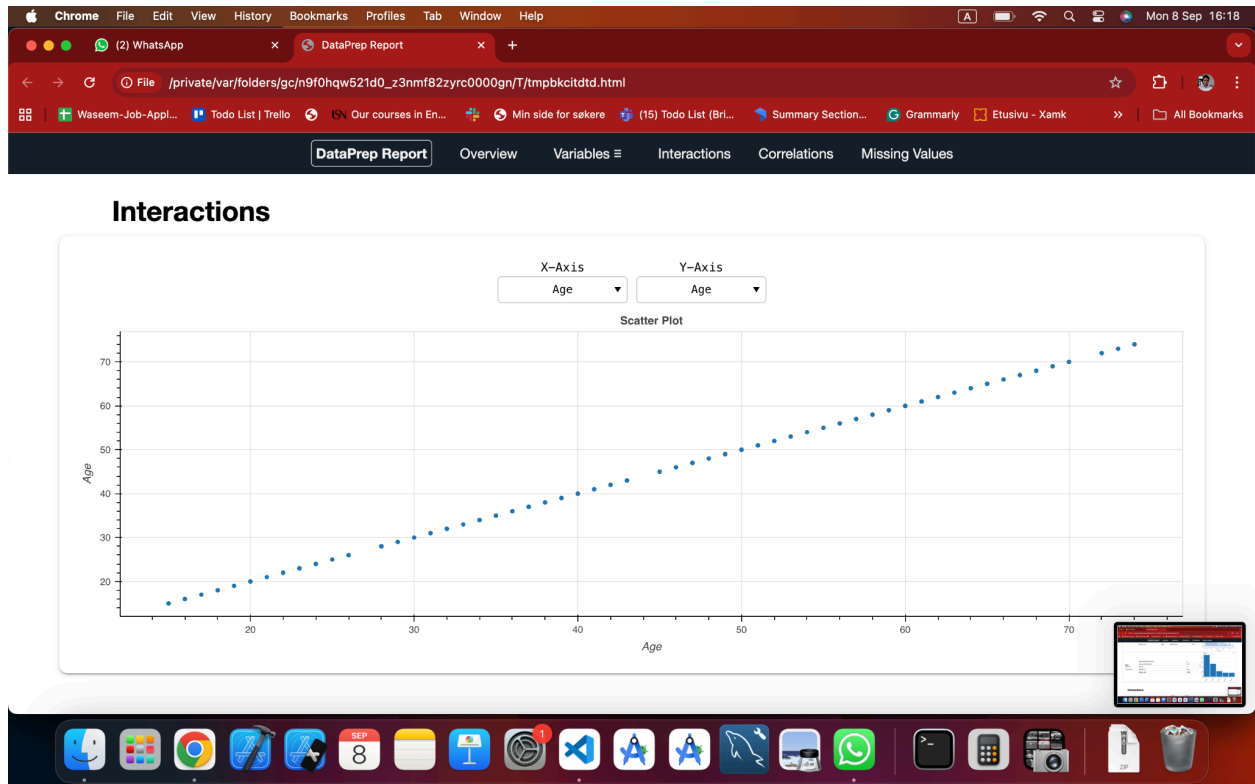


Cholesterol and Na_to_K

is a categorical variable with 2 distinct values and 1.0% approximate uniqueness.

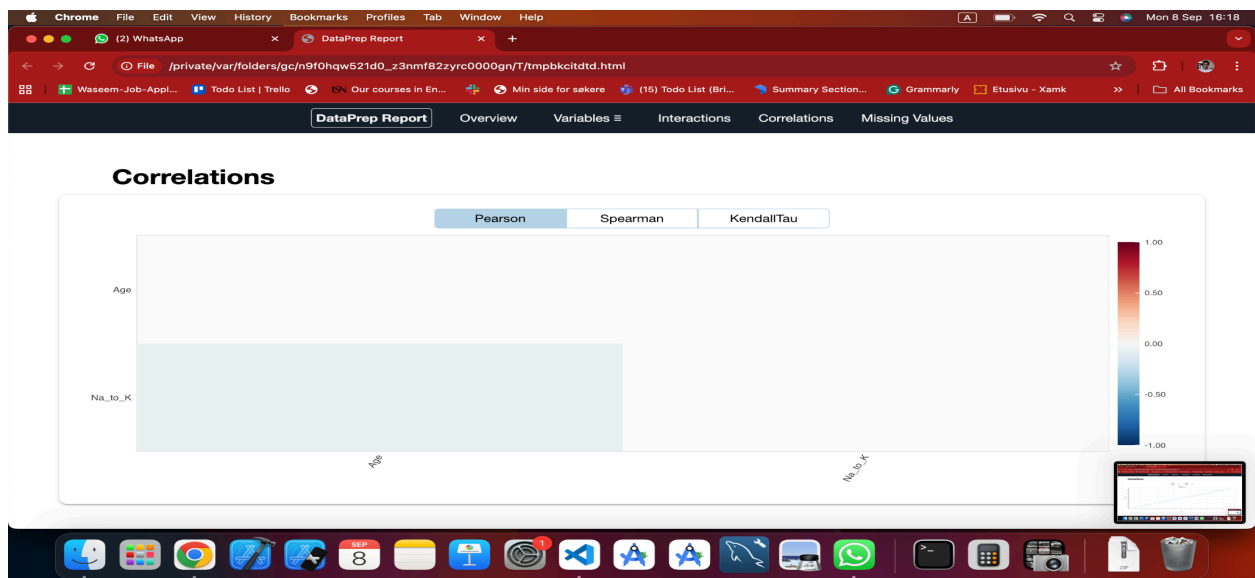
It has no missing data and occupies 13,994 bytes of memory. The variable Na_to_K has 198 distinct values, mean 16.0845, minimum 6.269, and maximum 38.247. There are no missing, zero, or negative values.

The distribution is right-skewed, with most values between 6–20. Memory size used is 3200.



Interactions

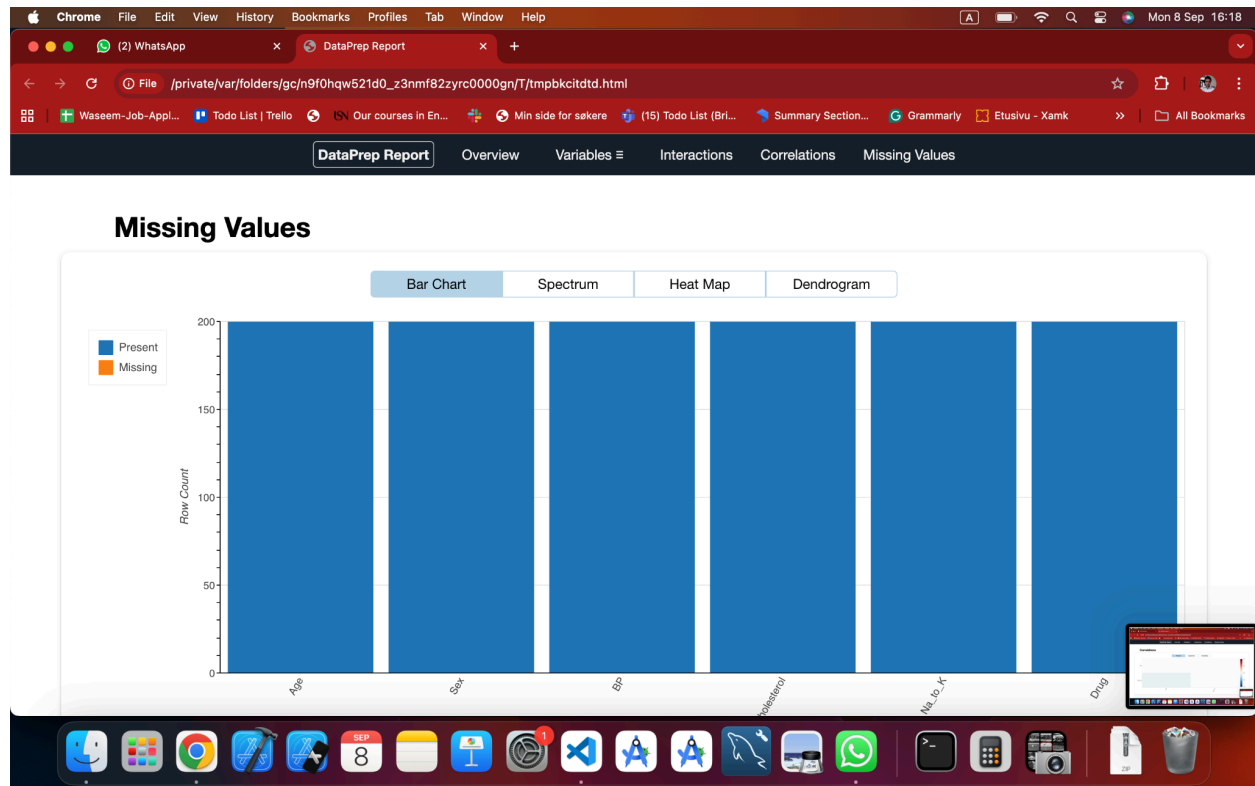
The first plot is Age against Age, so the line goes straight from 15 to 75 because both sides are the same. The second plot is Na_to_K against Na_to_K, making a straight line from about 6.2 to 38.2 for the same reason. In both cases, the x and y values match each other exactly.



Correlations between 2 variables

This heatmap visualizes the Pearson correlation between two variables: Age and Na_to_K (sodium-to-potassium ratio). The correlation value appears to be close to 0, as indicated by the very light

color in the matrix, meaning no significant linear relationship exists between these variables. The color bar on the right ranges from -1.0 (dark blue) to +1.0 (dark red), but the actual cell between Age and Na_to_K is nearly white, reinforcing the near-zero correlation. The selected method is Pearson, which measures linear relationships. No meaningful correlation pattern is present here. Exploring Spearman or KendallTau might reveal nonlinear or rank-based associations.



Missing values

There are different types of graphs, line heatmap, spectrun, dendrogram, but I am taking it as a bar chart showing the presence of missing values across various columns in a dataset. The chart visualizes the row count for each feature:

X-axis: Represents the dataset columns: Age, Sex, BP, Cholesterol, Na_to_K, Drug.

Y-axis: Represents the row count for each feature, showing the number of entries that either have present data (blue bars) or are missing data (orange bars).

All bars are blue, indicating that there are no missing values in any of the columns.

In conclusion, the dataset has complete data for all the features listed, with no missing entries in any column.

There are other images related to it also placed in the images folder for further images, Assignment01 images, then all drug images are classificaitons but insuarance have regression related, but each folder has a name showing a different library which is being produced by it.

Phik

```
In [2]: drug = pd.read_csv("drug.csv")
drug.head()
```

```
Out[2]:
```

	Age	Sex	BP	Cholesterol	Na_to_K	Drug
0	23	F	HIGH	HIGH	25.355	DrugY
1	47	M	LOW	HIGH	13.093	drugC
2	47	M	LOW	HIGH	10.114	drugC
3	28	F	NORMAL	HIGH	7.798	drugX
4	61	F	LOW	HIGH	18.043	DrugY

```
In [3]: import phik

phik_matrix = drug.phik_matrix()
phik_matrix
```

interval columns not set, guessing: ['Age', 'Na_to_K']

```
Out[3]:
```

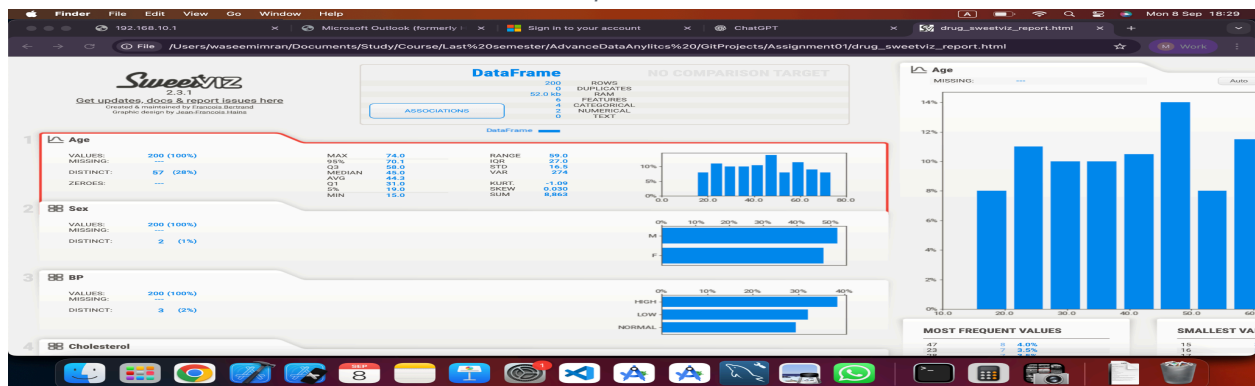
	Age	Sex	BP	Cholesterol	Na_to_K	Drug
Age	1.000000	0.012005	0.044947	0.202679	0.266617	0.366953
Sex	0.012005	1.000000	0.000000	0.000000	0.000000	0.000000
BP	0.044947	0.000000	1.000000	0.065489	0.000000	0.627489
Cholesterol	0.202679	0.000000	0.065489	1.000000	0.000000	0.230950
Na_to_K	0.266617	0.000000	0.000000	0.000000	1.000000	0.783475
Drug	0.366953	0.000000	0.627489	0.230950	0.783475	1.000000

This output shows the Phi correlation matrix, which measures both linear and non-linear relationships between variables. The strongest correlation is between Na_to_K and Drug at 0.783475, followed by BP and Drug at 0.627489, suggesting a strong association. The weakest correlations are seen with Sex, which has a Phik score of 0.0 with most variables, indicating no relationship. Age shows moderate correlation with Na_to_K (0.266617) and Drug (0.366953). The highest value (1.0) is along the diagonal, indicating perfect self-correlation.

Sweetviz

Sweetviz is used to quickly create a detailed visual report that compares and explains the dataset. It will create a complete report which is going to explain with images and descriptions as well.

Not, There re alot of images in the images folder and also descriptions with images in the file in coding comments. I will share a few of them that seem important.

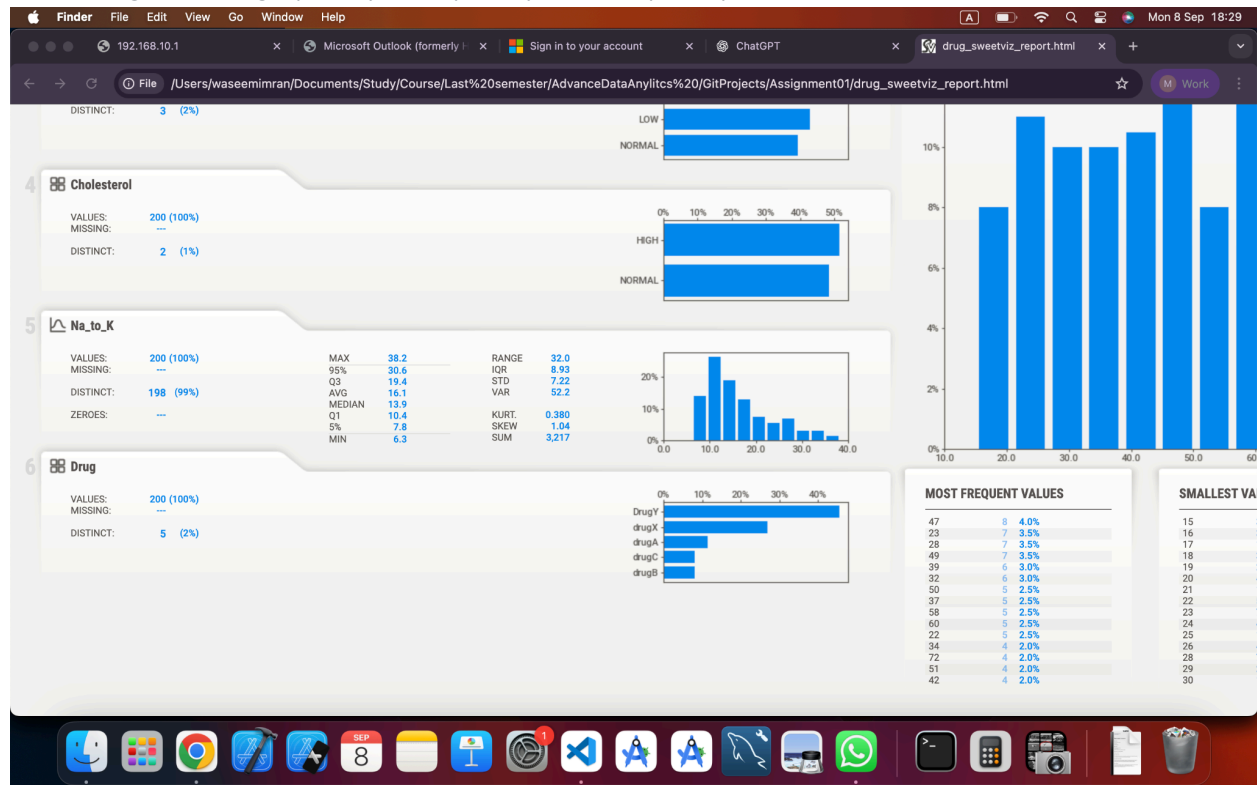


The dataset has 200 rows, 6 features (4 categorical, 2 numerical), and no missing values.

Age: Min 15, Max 74, Avg 44.3, Median 45, STD 16.5, Distinct 57 (28%).

Sex: 2 categories (M, F) ~50% each.

BP: 3 categories – High (~40%), Low (~30%), Normal (~30%)



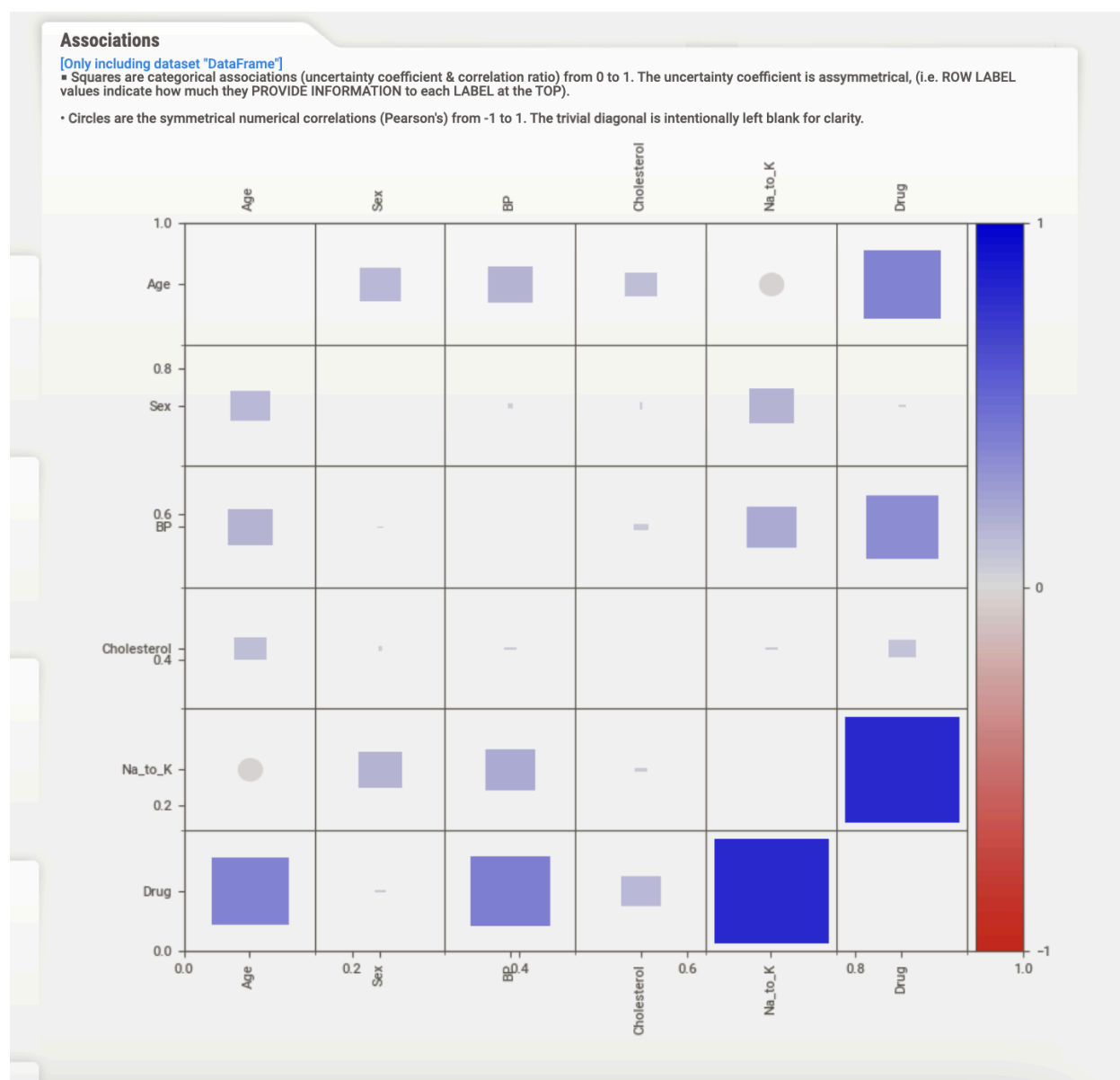
Cholesterol: 2 categories – High, Normal (~50% each).

Na_to_K: Min 6.3, Max 38.2, Avg 16.1, Median 13.9, STD 7.22, Distinct 198 (99%).

Drug: 5 categories – DrugY (~40%), DrugX, DrugA, DrugC, DrugB (smaller shares).

Distribution: Age is fairly spread, Na_to_K skewed right, and categorical variables are well balanced.

Overall: demographic features are balanced, but drug choice seems influenced by Na_to_K and BP.



Association Report with SwetViz

The plot shows correlations and associations between variables: age, sex, bmi, children, smoker, region, and charges.

Circles represent numerical correlations (Pearson's, from -1 to 1); squares represent categorical associations (uncertainty coefficient/correlation ratio, 0-1).

smoker and charges have the strongest association (~0.9), shown as a large dark blue square.

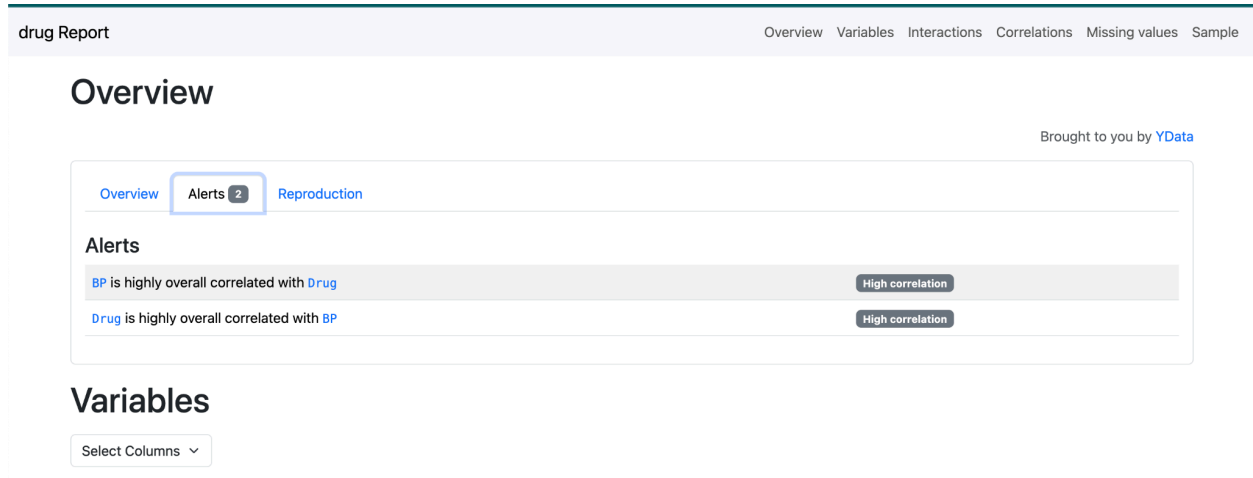
age and charges have a moderate positive correlation (~0.6), shown by a large blue circle.

BMI and charges also correlate moderately (~0.4).

Other variables (sex, children, region) show weak or negligible associations (<0.2) with charges.

YData Profiling

YData Profiling is used to automatically create a detailed report that summarizes and explains a dataset. Its mean we are going to experience another library with the same purpose.



This drug Report (generated by pandas-profiling/ProfileReport) shows 2 alerts: BP is highly overall correlated with the drug, and the drug is highly overall correlated with BP.

Note: Here I am not putting images, because it will become a very long file, Thereby Reprot (HTML file) is already there with the name of “Drug_Report.Html”, but I am putting description with variable names

This ProfileReport overview shows the dataset has 6 variables and 200 observations. There are 0 missing cells (0.0%) and 0 duplicate rows (0.0%). The dataset size is 50.8 KiB with an average record size of 259.9 B. Variable types: 2 Numeric and 4 Categorical.

Age is a numeric variable with values ranging from 15 to 74, a mean of 44.3, and 57 distinct values (28.5%) with no missing or zero entries.

The BP (Blood Pressure) variable is categorical with 3 levels: HIGH (77), LOW (64), and NORMAL (59), and no missing values.

Age distribution is fairly spread across the range with no negatives or infinities, memory size 1.7 KiB.

BP has balanced category counts, a mean string length of 4.27, and a memory size of 12.1 KiB.

The scatter plot shows Age vs Na_to_K. The Age values range from about 15 to 75, while Na_to_K values range from about 6 to 38. The points are spread out randomly, showing no strong linear relationship between the two variables. This means Age and Na_to_K do not directly move together.

The scatter plot shows Na_to_K vs Na_to_K. The values range from about 6.2 to 38.2, and all points lie perfectly on a diagonal line. This happens because the variable is plotted against itself, giving a one-to-one identity relationship.

or in other words All points form a perfect diagonal line because each Na_to_K value is exactly equal to itself. This confirms a one-to-one identity relationship between the two axes.

The strongest positive correlation is between Na_to_K and Drug (~0.75), meaning higher sodium-to-potassium levels are strongly linked with the drug type.

Age and BP (~0.40), and BP and Cholesterol (~0.45) also show moderate positive relationships.

Cholesterol and Drug (~0.30), as well as Age and Na_to_K (~0.25), indicate weaker positive links.

Sex has very low correlation with most variables, close to 0.0, meaning little to no linear relationship. There is a slight negative correlation between Age and Sex (~-0.10) and between Sex and Drug (~-0.05).

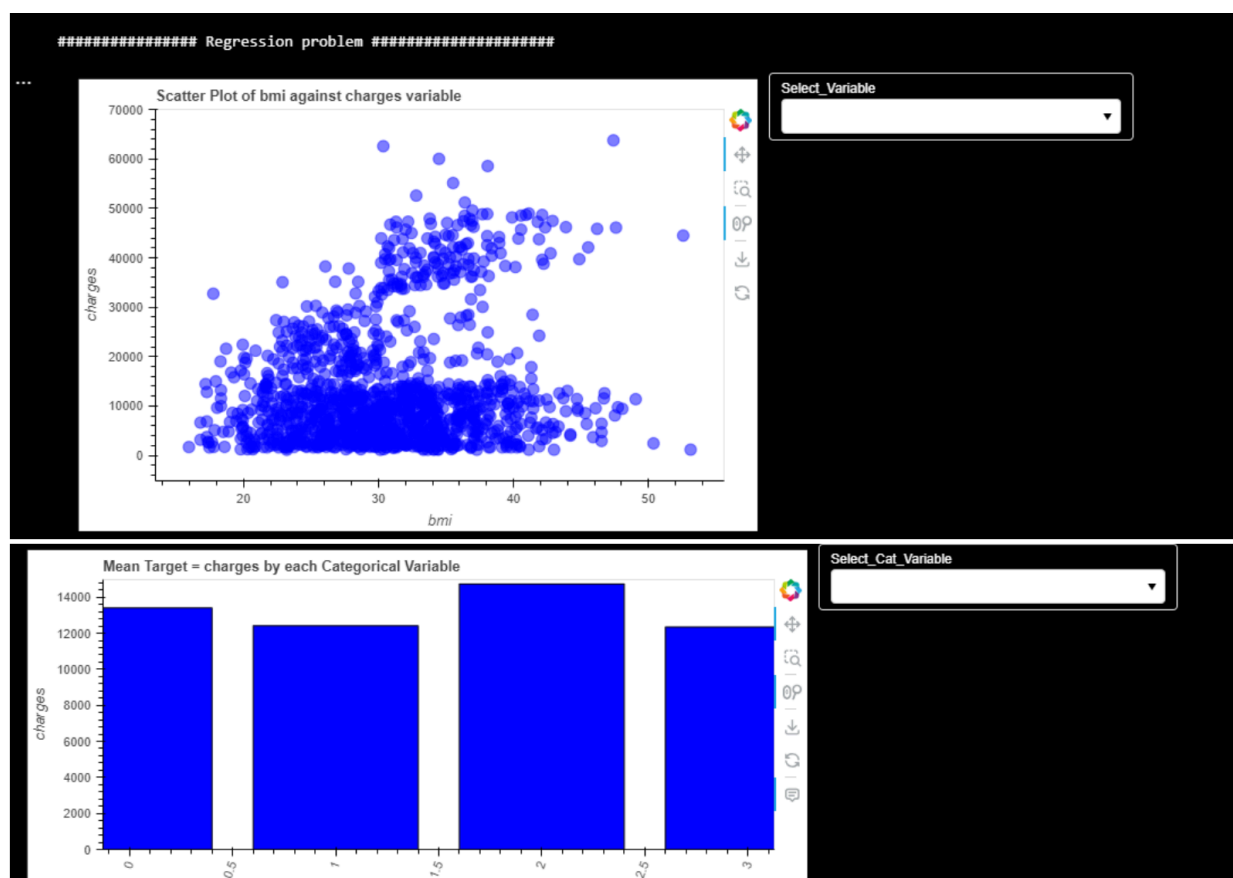
Diagonal values are 1.00 because each variable is perfectly correlated with itself.

Summary of it.

This dataset has 6 columns: Age, Sex, BP, Cholesterol, Na_to_K, and Drug. For example, a 23-year-old female (BP=HIGH, Cholesterol=HIGH, Na_to_K=25.355) → DrugY, and a 47-year-old male (BP=LOW, Cholesterol=HIGH, Na_to_K=13.093) → drugC. Drugs include DrugY, drugC, and drugX, given based on conditions. A 61-year-old female (LOW BP, HIGH Cholesterol, Na_to_K=18.043) → DrugY, while a 22-year-old female (NORMAL BP, HIGH Cholesterol, Na_to_K=8.607) → drugX. Overall, the report studies patient counts per drug, gender split, BP & Cholesterol levels, average Age/Na_to_K per drug, and how these factors affect which drug is prescribed most often.

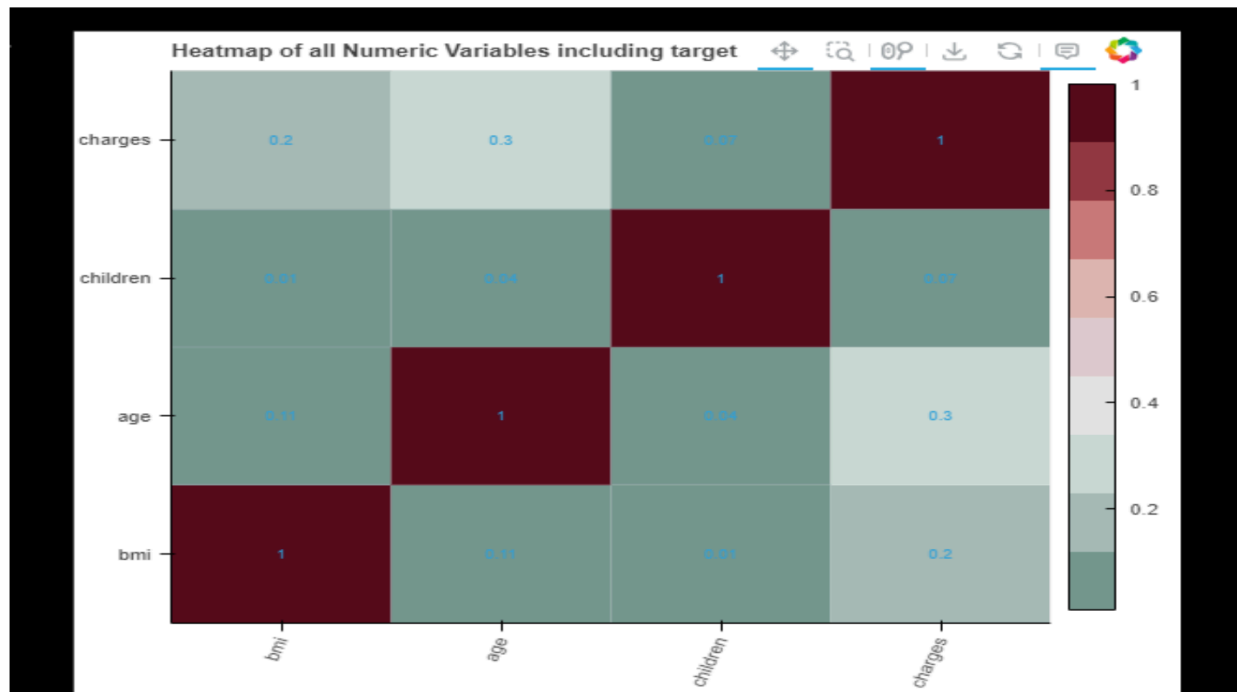
***Lets Talk about Regression
with the dataset and different
experiments with different
libraries, Same lib that we
applied at Classification
approaches.***

AutoViz

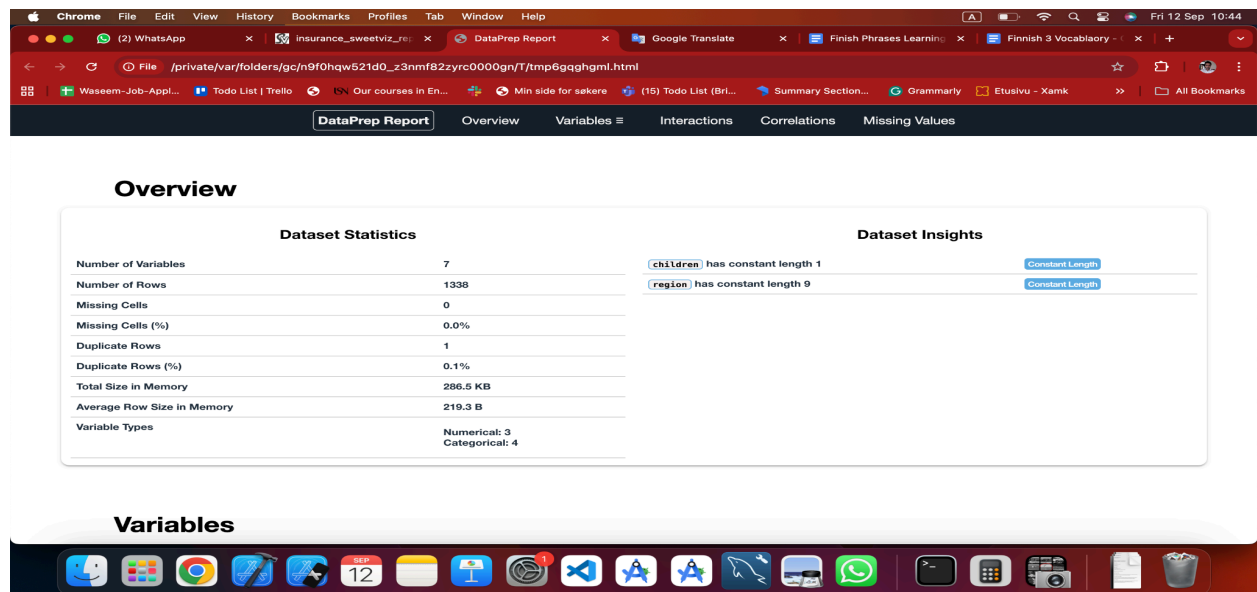


This scatter plot shows the relationship between **BMI** (ranging approximately from 15 to 55) and **insurance charges** (ranging roughly from 1,000 to 65,000). The plot highlights that **charges increase sharply for higher BMI values**, especially above **BMI ≈ 30** . The **mean charges** are significantly higher for specific categorical groups (notably smokers), where average charges exceed **30,000**, compared to non-smokers whose mean charges remain below **15,000**.

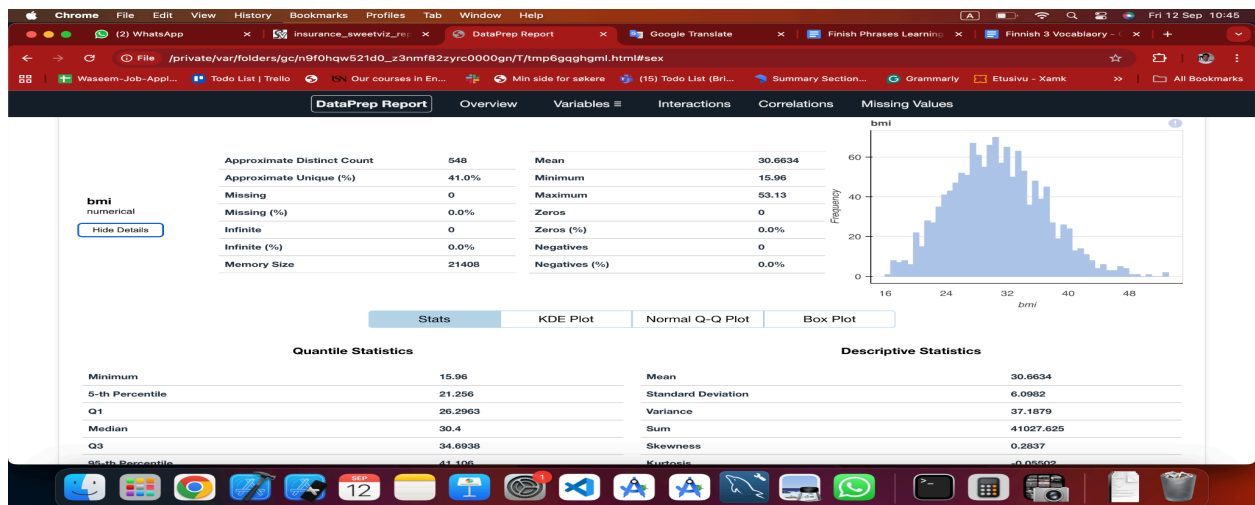
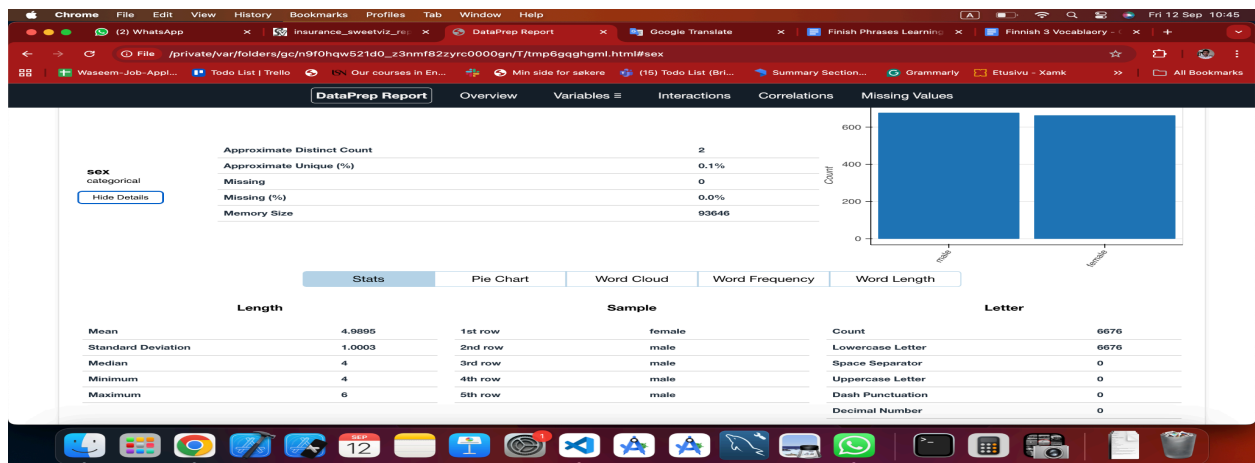
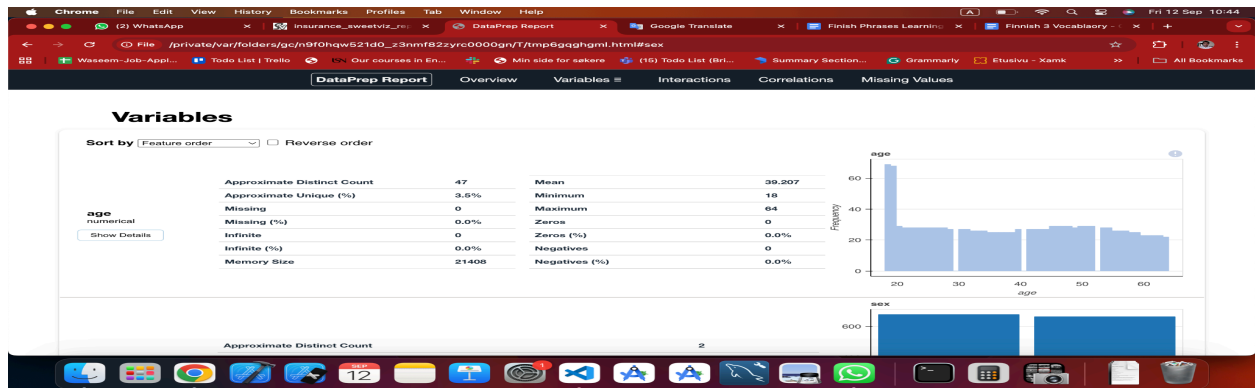
There are so many other graphs and explanations that I put in the notebook already. Images are in the image folder, but I picked 2 of the best images for explanation; otherwise, this file will be too long.



This heatmap displays the correlation coefficients among numerical variables such as **age**, **BMI**, **children**, and **charges**. The strongest positive correlation is observed between **age** and **charges** ($\approx 0.30\text{--}0.35$), indicating that insurance charges increase with age. **BMI** and **charges** show a weaker positive correlation (≈ 0.20), while **children** and **charges** have a very low correlation (≈ 0.07), suggesting minimal impact. The diagonal values are **1.00**, representing perfect self-correlation.



The dataset contains 7 variables and 1,338 rows with no missing cells and only 1 duplicate row (0.1%). It has a total memory size of 286.5 KB, with 3 numerical and 4 categorical variables. Additionally, the columns children and region have constant string lengths of 1 and 9, respectively.



The DataPrep library was used to analyze the dataset and visualize variable statistics. The age column (numerical) has 47 distinct values ranging from 18 to 64, with a mean of 39.21 and no missing or zero values. The sex column (categorical) contains 2 categories — male and female — almost equally distributed, with 0% missing data and a memory size of 93.6 KB. All characters in the sex column are lowercase, with no punctuation or numbers detected. This report highlights clean and well-distributed data.

bmi, children , smoker and BMI

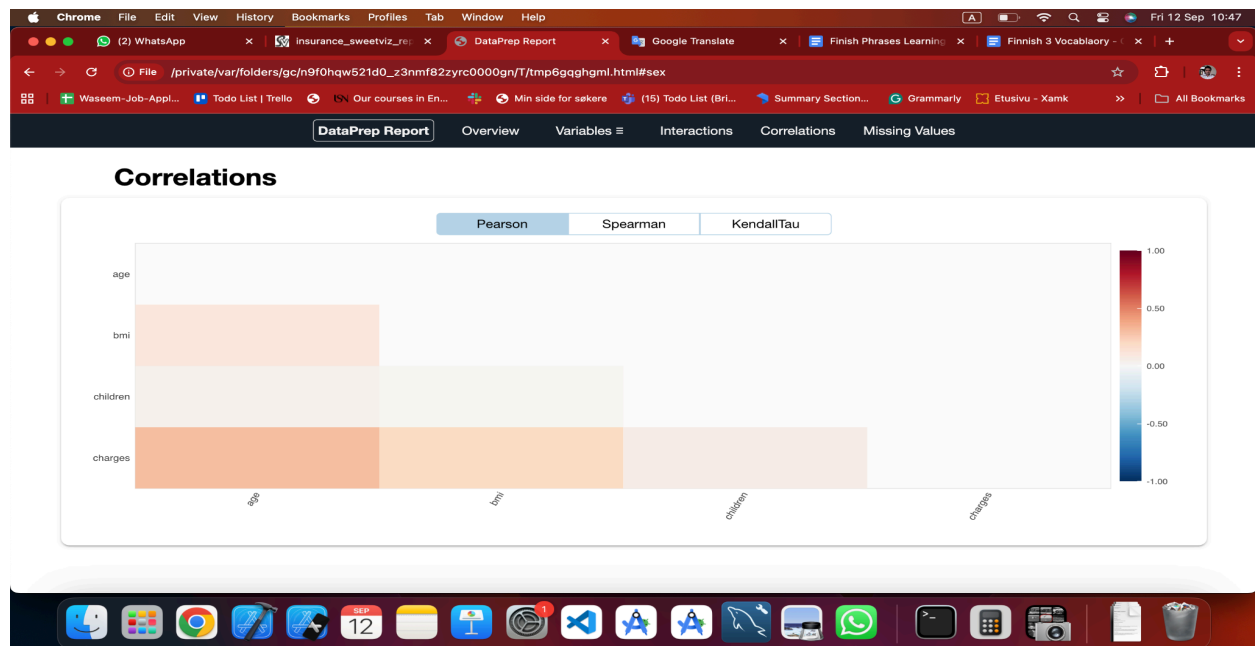
Using Dataprep, bmi ranges from 15.96 to 53.13, with a mean of 30.66 and median 30.4. The distribution is slightly right-skewed, with most values between 26.29 (Q1) and 34.69 (Q3). No missing or zero values are present, and the histogram shows a normal-like spread.

The children variable has 6 categories (0–5), with most people having 0 or 1 child. Over 500 records show 0 children, while counts decrease as the number of children increases.

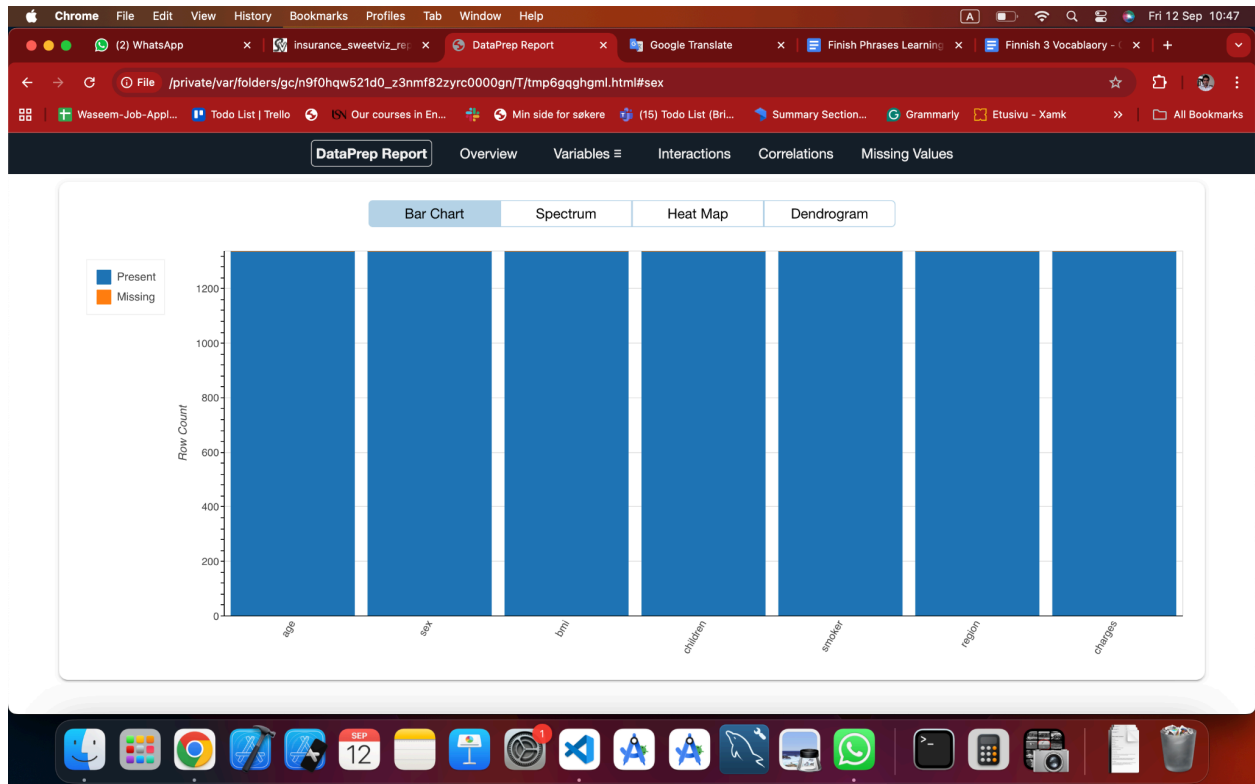
No missing values exist, and the distribution is highly skewed toward smaller family sizes.

The smoker variable has 2 categories: yes and no. you make call it boolean in terms of languages. Most people are non-smokers (~1060), while around 270 are smokers. No missing values, and the chart shows a clear imbalance toward the non-smoker group.

The scatter plot shows BMI vs BMI, forming a perfect straight line. BMI values lie between 15 and 53. Every point falls on the line $y = x$, meaning correlation = 1.0. This happens because a variable compared to itself is always identical.

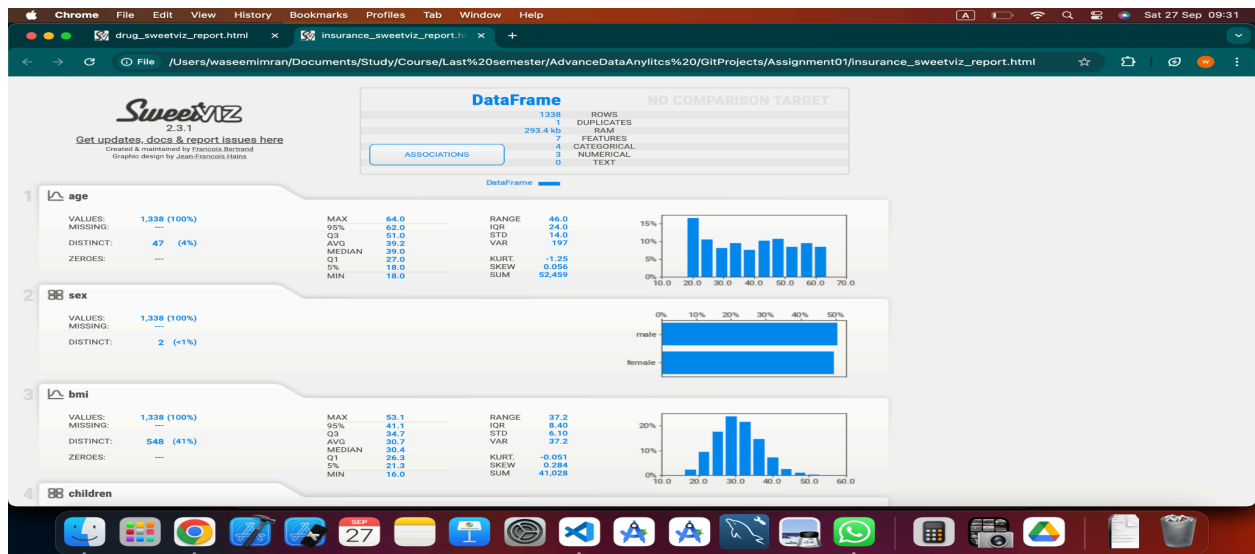


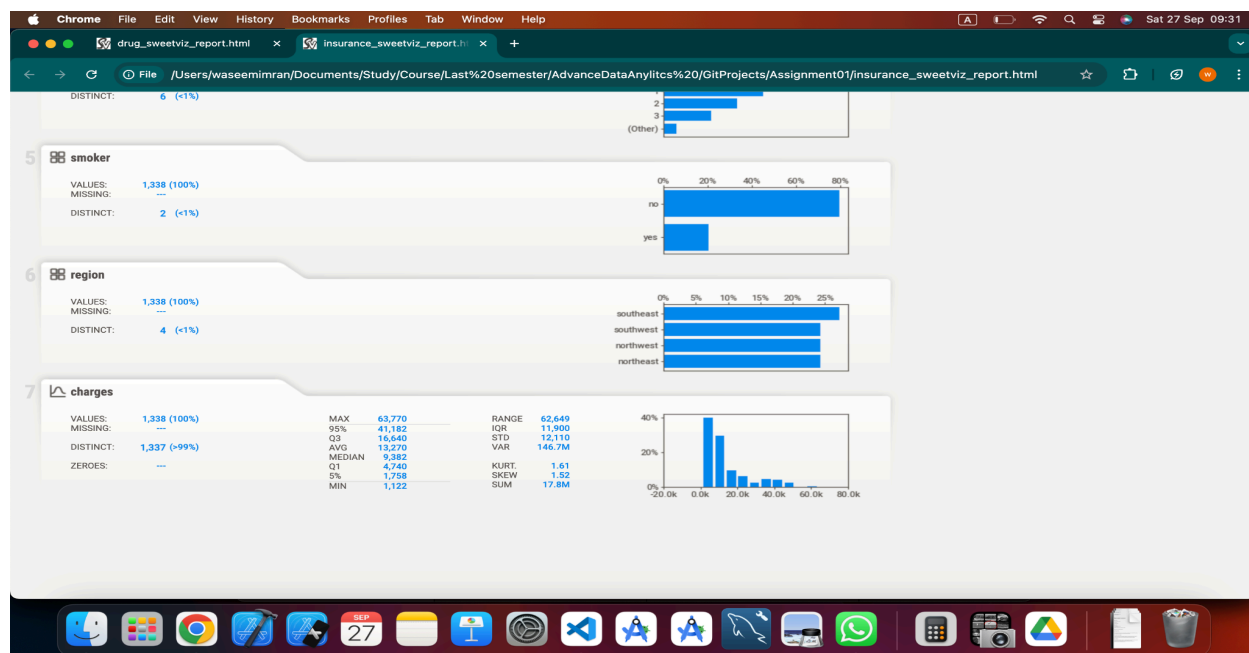
The Pearson correlation heatmap shows that charges have a moderate positive correlation with age (~0.3) and BMI (~0.2), meaning older individuals and those with higher BMI tend to incur higher charges. Charges also show a weaker positive correlation with children (~0.1). Age and BMI themselves are weakly correlated (~0.1), while children show almost no correlation with age or BMI (close to 0). Overall, charges are most influenced by age and BMI, but the relationships are not very strong.



his bar chart shows all variables in the dataset. Each column (age, sex, bmi, children, smoker, region, charges) is completely filled with blue, which represents “Present” values. There is no orange portion, meaning 0 missing values in any of the variables. The row count for each variable is around 1338 records.

Sweetviz





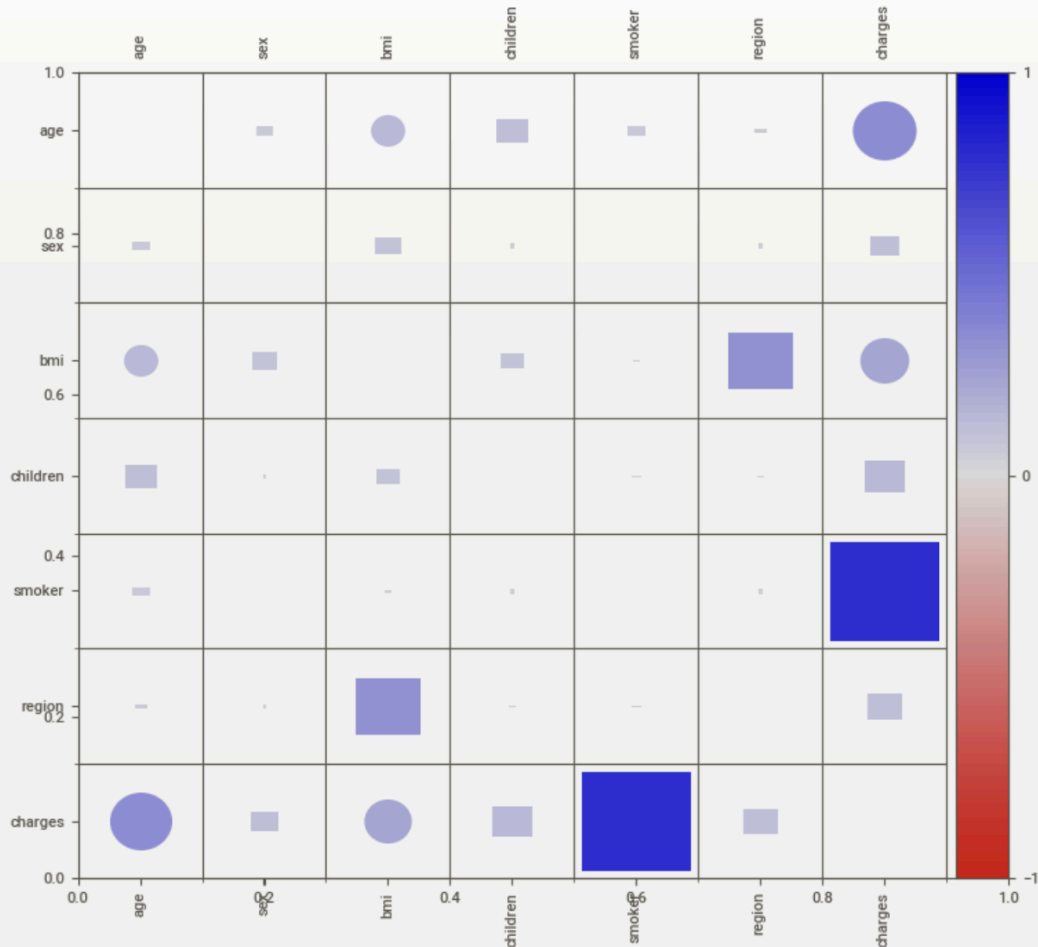
The insurance dataset has about 1300 rows with features like age, sex, BMI, children, smoker status, region, and charges, and it contains no missing values, making it clean to use. Charges, the main variable, are skewed, with most people paying moderate amounts and a few paying very high. Age ranges from young adults to older people, BMI goes from healthy to obese, and children count varies across families. Categorical features such as sex, smoker, and region are evenly distributed, though there are more non-smokers than smokers.

Associations

[Only including dataset "DataFrame"]

• Squares are categorical associations (uncertainty coefficient & correlation ratio) from 0 to 1. The uncertainty coefficient is asymmetrical, (i.e. ROW LABEL values indicate how much they PROVIDE INFORMATION to each LABEL at the TOP).

• Circles are the symmetrical numerical correlations (Pearson's) from -1 to 1. The trivial diagonal is intentionally left blank for clarity.



The report shows that smokers have much higher insurance charges than non-smokers, making smoking the strongest driver of cost. Age and BMI also increase charges but not as strongly, while the number of children has only a small effect, and region or sex matter very little. A few people with very high BMI, age, or charges appear as outliers compared to the rest. Since the dataset is complete, no cleaning is needed, and analysis can focus on using key predictors like smoker status, age, and BMI, as well as grouping people by smoking habits, age ranges, or BMI levels for better insights.

The association matrix shows that **smoker status has the strongest relationship with charges (≈ 0.79)**, making it the most influential factor. **Age has a moderate positive correlation with charges (≈ 0.30)**, followed by **BMI, which also shows a moderate association (≈ 0.20 – 0.25)**. **Number of children has only a weak association with charges (≈ 0.10)**, contributing little explanatory power. **Sex (≈ 0.06) and region (≈ 0.05)** have negligible associations with charges, indicating minimal impact.

YData Profiling

The screenshot shows the YData Profiling interface for an 'insurance Report'. The 'Overview' tab is selected, and the 'Alerts' sub-tab is active, showing 5 alerts. The alerts are:

- Dataset has 1 (0.1%) **duplicate rows** (button: Duplicates)
- age** is highly overall correlated with **charges** (button: High correlation)
- charges** is highly overall correlated with **age** and 1 other fields (button: High correlation)
- smoker** is highly overall correlated with **charges** (button: High correlation)
- children** has 574 (42.9%) zeros (button: Zeros)

The 'Variables' section below shows a 'Select Columns' dropdown.

The screenshot shows the YData Profiling interface for an 'insurance Report'. The 'Overview' tab is selected, and the 'Overview' sub-tab is active. The page displays 'Dataset statistics' and 'Variable types'.

Dataset statistics	
Number of variables	7
Number of observations	1338
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	1
Duplicate rows (%)	0.1%
Total size in memory	286.5 KiB
Average record size in memory	219.3 B

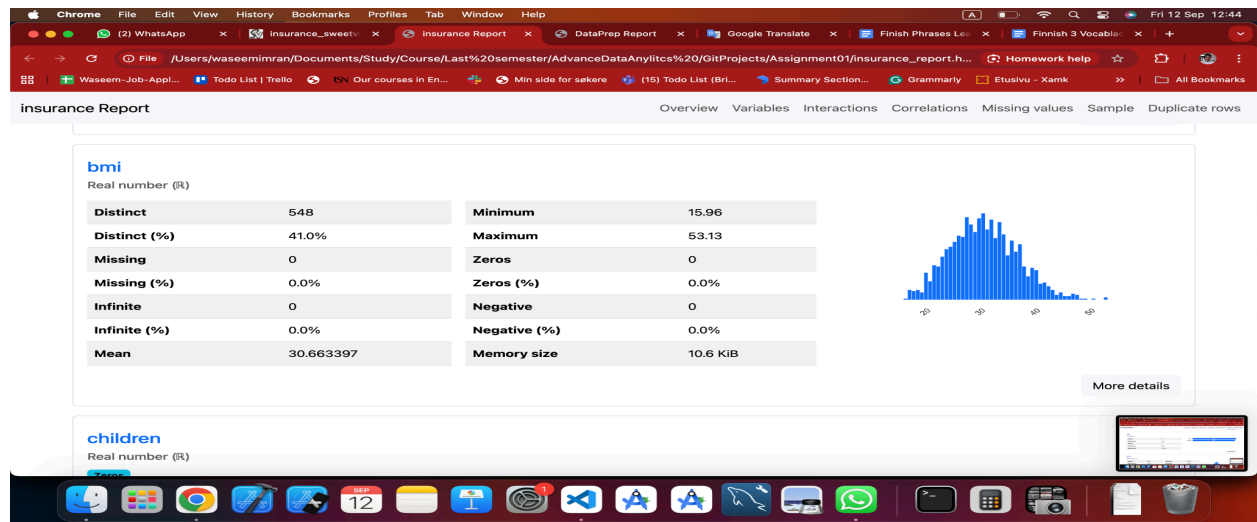
Variable types	
Numeric	4
Categorical	2
Boolean	1

Duplicates: The dataset contains 1 duplicate row (0.1%), which may slightly affect analysis accuracy.
High Correlations: age is highly correlated with charges; charges is also highly correlated with age and 1 other field; smoker is highly correlated with charges.
Zeros: The children column has 574 zeros, representing 42.9% of the dataset, indicating many entries with no children.

The dataset has 7 variables and 1338 observations with no missing values, ensuring data completeness. Only one duplicate row exists, which is about 0.1% of the dataset. The entire dataset takes up 286.5 KB

in memory, with each record averaging 219.3 bytes. Among the variables, 4 are numeric, 2 are categorical, and 1 is boolean.

All Variable images in image folder with profile insurance name, So, I am putting here just only description if i put all images then this file will be too long.



BMI (Body Mass Index):

BMI ranges from 15.96 to 53.13 with a mean of around 30.66, showing overweight on average.

No missing, zero, or negative values.

The histogram shows a normal-like distribution, centered around 25–35.

Charges:

Charges vary widely, from about 1,121 to 63,770, with a mean of 13,270.

Almost all values are unique (99.9% distinct) and no missing data.

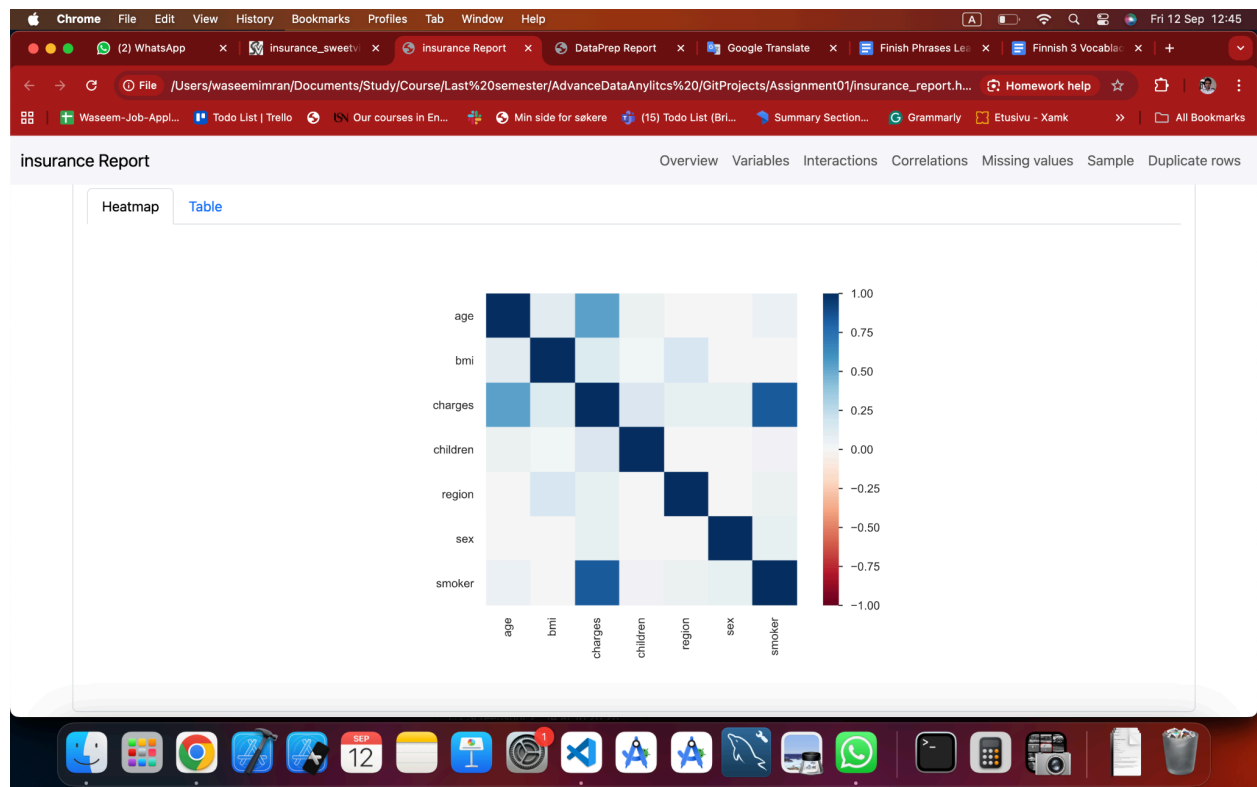
Distribution is right-skewed, with most charges clustered at lower values and a few very high outliers.

Children:

Number of children ranges from 0 to 5, with a mean of about 1.1.

Many people have 0 children (42.9%), making zeros the most common value.

The histogram shows decreasing counts as the number of children increases.



Age and Charges have a strong positive correlation ($\sim 0.3-0.4$), meaning older people tend to pay higher insurance charges.

Smoker and Charges show the strongest correlation ($\sim 0.75-0.8$) — smoking greatly increases medical costs.

BMI and Charges also correlate positively ($\sim 0.2-0.3$), so higher BMI is linked with higher charges.

Children, Region, and Sex show very weak correlations (close to 0), meaning they hardly affect charges directly.

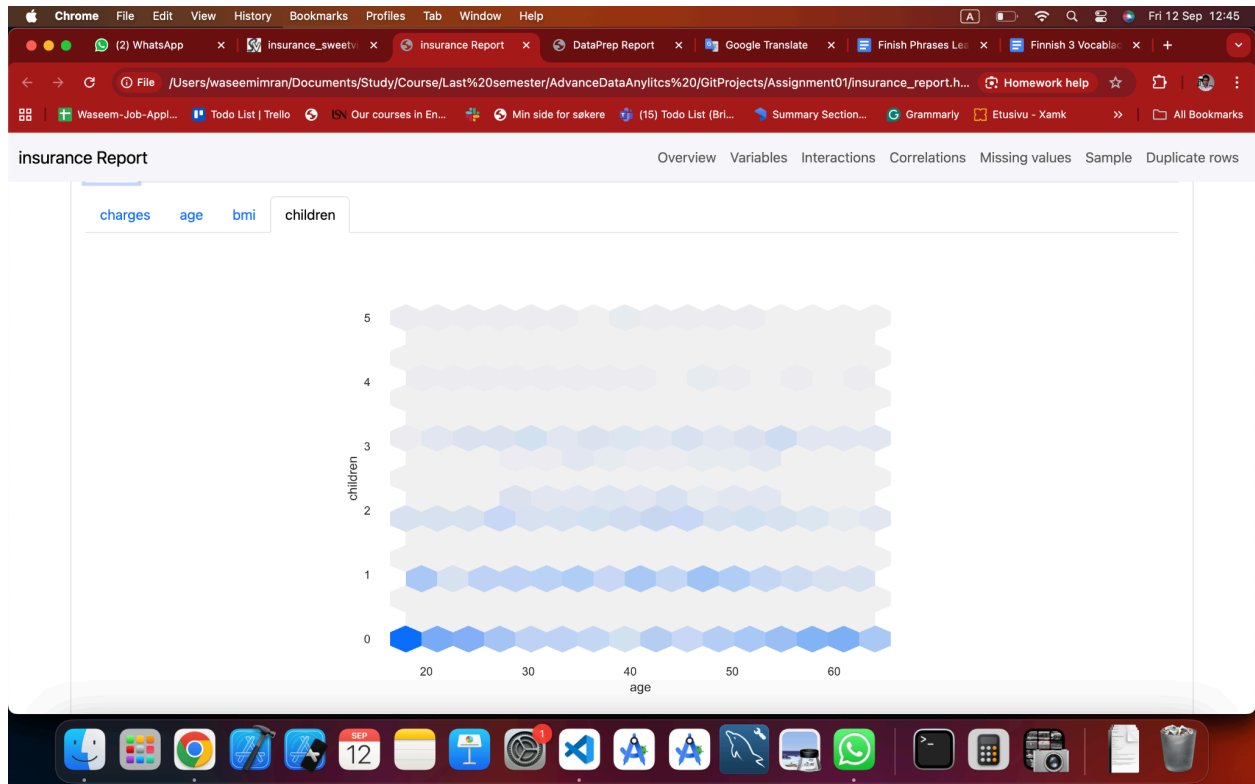
Diagonal values are all 1.0, since each variable is perfectly correlated with itself.

The heatmap uses blue for positive and red for negative correlations, but here most are weakly positive.

Smoker (≈ 0.8) → Being a smoker has the largest impact on insurance charges. Smokers pay significantly higher costs.

Age ($\approx 0.3-0.4$) → Older individuals are likely to have higher medical expenses, so age is the second strongest factor.

BMI ($\approx 0.2-0.3$) → Higher body mass index (BMI) moderately increases charges, linking obesity with higher medical risk.



This chart shows the relationship between age and number of children. Most people across all age groups have 0 to 2 children, with a higher density at younger ages having no children. As age increases, the distribution spreads, but still, very few people have more than 3–5 children. Overall, there is no strong direct correlation between age and number of children in this dataset.

Phik

```
In [1]: # importing the usual needed modules
import pandas as pd

In [2]: insurance = pd.read_csv("insurance.csv")
insurance.head()

Out[2]:
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

This phik correlation matrix shows the strength of associations between variables (0 = no relation, 1 = perfect relation). In summary:

Charges are strongly related to smoker (0.96) and age (0.72), with a moderate link to BMI (0.38). Age and children (0.32) show some association, while most other pairs are very weak or near zero. Sex and region have little to no meaningful correlation with most variables.

Overall, smoking status is the dominant factor linked to charges, followed by age and BMI, while sex and region barely matter.

```
In [3]: import phik

phik_matrix = insurance.phik_matrix()
phik_matrix
```

interval columns not set, guessing: ['age', 'bmi', 'children', 'charges']

```
Out[3]:
```

	age	sex	bmi	children	smoker	region	charges
age	1.000000	0.000000	0.155935	0.316187	0.000000	0.000000	0.724628
sex	0.000000	1.000000	0.000000	0.000000	0.108406	0.000000	0.081934
bmi	0.155935	0.000000	1.000000	0.086769	0.000000	0.269831	0.377421
children	0.316187	0.000000	0.086769	1.000000	0.052318	0.000000	0.033286
smoker	0.000000	0.108406	0.000000	0.052318	1.000000	0.086139	0.962946
region	0.000000	0.000000	0.269831	0.000000	0.086139	1.000000	0.108148
charges	0.724628	0.081934	0.377421	0.033286	0.962946	0.108148	1.000000

Charges (insurance cost) is strongly correlated with smoking (0.96) and age (0.72), meaning smokers and older people pay much higher charges.

BMI also has a moderate positive correlation with charges (0.38), so higher BMI tends to increase costs.

Children, sex, and region show very weak or no correlation with charges.